

# 나현준 (Hyun June Na)

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## EDUCATION

Seoul National University, B.S. in Mechanical Engineering

2022.03 – Present

## PROJECTS

**CNSL Screw Press — Waste-to-Value Oil Extraction Machine (Team COMET)** 2026.03 – 2026.06

*Interdisciplinary Capstone Design, Dept. of Mechanical Engineering, SNU (Advisor: Prof. Sung-Hoon Ahn) — Team Lead (4-member team)*

- Led a 4-member capstone team to design and build a screw-type press extracting CNSL (cashew nut shell liquid), a phenolic industrial oil, from waste cashew shells for low-infrastructure use in Tanzania
- Designed the full machine in 3D CAD (Fusion 360, SolidWorks) with shell-specific features — variable-pitch screw, tapered discharge slits, barrel protrusions, and poppet-valve pressure regulation — optimized via an oil-flow model built on cashew-shell material properties
- Selected the geared motor based on calculated extraction pressure and target oil-flow requirements
- Drafted ~10 2D fabrication drawings (AutoCAD) and managed the full vendor cycle — quoting, drawing revisions with fabricators, and part delivery — while sourcing standard components (geared motor, couplings, bearings) to spec, within a ~KRW 5M budget
- Led full mechanical assembly in a single overnight session, resolving dimensional mismatches and on-site part rework with hand and machine tools; verified oil extraction using substitute feedstocks (peanut, sesame) matched to cashew-shell oil content, viscosity, and hardness
- Selected for the 2026 Appropriate Technology-based Youth Startup Global PoC Program (Tanzania field pilot) and advanced to the finals of the S.M.A.R.T International Sustainable Technology Competition

**WAM (Weightlifting Assistant Machine) — Clamp-on Strength Training Assist Device** 2026.03 – 2026.06

*Mechatronics Course Team Project (Team 10, 5 members) — Hardware Lead*

- Led mechanical design and fabrication of a clamp-on gym-machine assist device, owning the full hardware path from concept to working prototype
- Redesigned the drive mechanism from a rack-and-pinion actuator to a drum-and-strap cable-winding system to cut weight and enable attach/detach on existing machines
- Modeled the motor-drum-strap power transmission in CAD and fabricated the full enclosure and mechanical assembly
- Devised an adjustable clamp mount compatible with nearly all standard "plate & pin" gym machines, the key differentiator from single-machine built-in competitors
- Sized the system to a low-cost motor under budget, achieving stable 6 kg assistance on a self-built test rig with headroom to scale via a stronger motor
- Placed 4th of 16 teams in the course's final project competition

## AWARDS & SCHOLARSHIPS

- Advanced to the finals of the S.M.A.R.T International Sustainable Technology Competition (Academic Society for Appropriate Technology) with the CNSL screw press project
- Awarded 특상 for placing 4th of 16 teams in the Mechatronics course final project competition (WAM project)
- Selected for the 2026 Appropriate Technology-based Youth Startup Global PoC Support Program (Sanhak Foundation · UNITEF · Academic Society for Appropriate Technology) — one of 4 Tanzania-track teams (8 nationwide) for funded on-site validation of the CNSL screw press in Tanzania